



Looking for the Best Partners

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Abstract. Interaction and cooperation among agents are two fundamental activities having place in multi-agent systems. In this regard, in a generally large universe of agents, the main difficulty lies in determining which agents are the most promising candidates for achieving fruitful interaction (i.e., cooperation). In a holistic view, several strategies like trust, reliability, reputation, cyphering or commitment can play, for example, a crucial role in various online environments. They provide a multitude of mechanisms for assessing credibility, trustworthiness, identity and so on, not only with respect to agents but also considering entities such as users, products or services. Agent-based scenarios can take advantage from these techniques, which are also able to reduce information asymmetry between agents, enhancing their mutual confidence and, in this way, encouraging their participation by limiting the probability of deception. Moreover, such systems form the basis for a potentially large number of countermeasures to prevent, depotentiate or nullify malicious activities intended at altering a correct perception about the counterpart by means of misrepresentation or manipulation activities carried out by bad agents. In this context, the goal of this chapter is to provide an overview of the several specific proposals, made at Workshop on Objects and Agents (WOA) over more than two decades, aimed at identifying the best partners for interaction and cooperation based on the above-cited approaches.

Keywords: Multiagent System · Partnership · Trust · Reputation · Cyphering · Commitment

1 Introduction

In the digital age, virtual communities are essential in the execution of several activities, and all without the need of physical interactions [53]. This relational paradigm requires a strategic approach for choosing one or more partners to be engaged in interaction and collaboration tasks, regardless of their real or virtual nature [46]. In fact, in online contexts, such tasks are characterized by

a complex web of relationships and connections [76], and in the decision to establish collaborations, participate in transactions, or share resources, aspects such as comfort and trustworthiness play a key role in shaping the experience of participants in these activities [77].

In detail, in virtual communities the multi-agent systems (MASs) represent a cornerstone of distributed computing, in which individual agents can collaborate with each other to achieve common or individual goals [132]. This approach is inspired by the concept of distributed intelligence observed in biological and social systems, where interactions between independent entities lead to emergent and adaptive behaviors [79].

Specifically, an agent is an autonomous computational entity with the ability to perceive the environment where it is living [135], process information [128], and act autonomously [136] to achieve assigned or independently identified goals. In the particular context of MASs, social relationships between agents can occur directly, through communication between agents, or indirectly, through observation of the results of other actors' actions in the shared environment [134]. Such social interaction mechanisms often underlie learning, also known as social learning [91], that provides agents with a dynamic adaptivity to the environment and improve their strategies and skills over time [16]. For such reasons, MASs find application in a wide range of fields, from robotics [81] to finance [9], from logistics [56] to health care [73], and have significantly contributed to the way we conceive and implement computing solutions.

In this scenario, as in human activities, successful interactions and cooperative tasks are essential for achieving complex goals in MASs [133]. So that the choice of the partners with whom to carry out such activities is crucial to obtain goals that could not be achievable individually [45]. This decision can significantly influence the outcome of the agents, bringing with it a number of challenges and considerations that must be carefully evaluated for achieving effective results, reaching common goals and promoting individual and collective success [45]. However, agent activities in MASs can also include situations of conflict, competition, or maliciousness [44] in which agents compete for limited resources, for the achievement of conflicting goals, or simply to gain undue advantages to the detriment of other actors [60].

One of the main challenges in selecting the best agent partners for interaction (i.e., cooperation) is the wide range of parameters that must be considered to build meaningful relationships within the community, thereby strengthening social capital and creating a connective tissue that extends beyond individual activities [96]. This because each potential partner brings with it a (unique) set of skills and experience [78]. Furthermore, another crucial aspect is to assess the alignment between the expectations, goals, and operating modes of the actors involved as well as their reliability and credibility, aspects that can undermine confidence between the parties [60].

Therefore, choosing the best partners is not only an individual decision, but has broader implications for the virtual community on the whole [92]. Indeed, well-considered decisions based on reliable information contribute to building

a strong, resilient and sustainable community over time [60]; conversely, poor choices can undermine overall confidence and community cohesion.

In conclusion, the choice of the best partners is based on sophisticated relational dynamics, and the continuous evolution of virtual communities accentuates the awareness in the importance of building positive relationships. All this in the perspective that virtual communities become even more resilient, connected and capable of facing the emerging challenges of the digital society. The careful choice of the best partners thus becomes a foundation for the future of relationships having place both in real and virtual worlds.

The background described above is fully supported by scientific literature, spanning multiple research fields. In fact, finding the best partner for interaction and cooperation is a problem that cuts across various application domains. Solving this problem has always been a challenge for researchers, maintaining its appeal over time, as evidenced by the overwhelming number of scientific contributions that annually enrich the body of knowledge. Over the past thirty years, these proposals have become more and more sophisticated, and complex, keeping pace with the evolution of society and technology, as well as advancements in languages and tools. From this perspective, the scientific research presented over the first twenty-five years of WOA meetings of the Italian community of agents aligns closely with the above context. Indeed, the numerous studies presented at WOA over the years have factually enriched the scientific literature, contributing to the anticipation and expansion of trends that later became well-established in both academic and industrial fields. In particular, the theme of partner search has been addressed in a significant number of WOA's papers.

In this celebratory book, this chapter focuses on highlighting WOA's contributions to the topic of identifying the best candidates for interaction or cooperation. As previously mentioned, a significant number of WOA papers have addressed this issue, and these contributions can be categorized into several strategies encompassing all research proposals. In particular, we have identified key strategies that rely on mechanisms of trust and reputation, ciphering, and commitment. This selection is based on the ability of these approaches to reduce information asymmetry among agents by assessing credibility, trustworthiness, and identity. These strategies encourage participation, limit the probability of deception, and, complementarily, identify malicious agents through their past actions, their identification, or their accountability. This overview, although focused on a specific theme, provides an interesting snapshot of the activity carried out by WOA in the field of agents, thanks to the significant number of papers described.

Moreover, this chapter is linked to that of Sapienza and colleagues in [119]. Here, trust is considered a key social construct that significantly influences decision-making and relationships in human, artificial, and hybrid societies. Specifically, Sapienza et colleagues investigate the evolution of the concept of trust in multi-agent systems through a systematic analysis of studies presented at the WOA workshop. In particular, the aim of [119] is to explore how the literature on trust in MAS has evolved over time. The chapter emphasizes that as interactions become more complex, so does the importance of trust, requiring

a deeper investigation of its various forms and dimensions. Furthermore, building on both the issues that have been addressed and those that remain open, they aim to outline future research directions and the challenges that still need to be addressed.

The remaining of the chapter is organized as follows. Section 2 describes the adopted methodology to select the most interesting papers presented at the WOA workshops and in detail the contribution of this chapter. Section 3 deals with trust, reputation and reliability issues, while Sect. 4 focuses on cipher-based systems, and Sect. 5 describe some commitment based solutions. Finally, some conclusions are drawn in Sect. 6.

2 Methodology and Contribution

From the first edition of WOA, which took place in the year 2000, to the 25th in 2024, 515 regular papers have been presented. The selection process for identifying the most relevant papers on looking for the best partners for interaction and cooperation tasks involved two consecutive steps. The first step involved searching for a range of relevant keywords (e.g., trust, reliability, reputation, cryptography, commitment, partner, cooperation, etc.) across all the 515 regular WOA papers. In the second step, each of these pre-selected papers was evaluated not only based on the frequency of the keywords used but also on their consistency and relevance to the focus of this chapter. After this selection process 53 scientific contributions resulted.

Building on this foundation, the contribution of this chapter aims to provide a systematic overview of the theme of identifying the best partners for interaction and cooperation. To achieve this, the 53 selected papers have been organized into three main sections: *(i)* Trust and Reputation Systems; *(ii)* Ciphering Systems; and *(iii)* Commitment Systems. Particular emphasis has been placed by WOA's authors on Trust and Reputation Systems, which cover a broad range of fields. Consequently, this section has been further subdivided into additional

Table 1. Contributions for theme

Trust and Reputation Systems	WOA contributions
Models	[22, 51, 52, 62, 86, 105] [108, 113, 124, 130]
Social Networks	[34, 57, 63–65, 80] [38, 40, 82, 83, 87, 112]
Information, Knowledge and Resources	[10, 20, 32, 49, 50, 110] [120, 122, 123, 125]
Embodied Agents	[25–27, 30, 58]
e-Business	[14, 90, 121]
Transportation Systems	[97, 100, 101]
Ciphering systems	[3, 8, 13, 66, 99, 107] [1, 131]
Commitment Systems	[4, 6]

Table 2. Contribution for year

Year	WOA contributions	Year	WOA contributions
2000	[8]	2013	[1]
2001	[20]	2014	[13,100,112,123]
2002		2015	[30,124]
2003		2016	[32,40,120]
2004	[22,90,105,130]	2017	[6,34,49,63,65,87,97]
2005	[10,107]	2018	[50,64,80,125]
2006		2019	[3,25,57,101,121]
2008		2020	[86]
2007	[99]	2021	[27,58]
2009	[14]	2022	[51,52,82,108]
2010	[131]	2023	[26,83]
2011	[110,113]	2024	[62,122]
2012	[4,38,66]		

subsections. Table 1 and Table 2 present the selected papers, organized by theme and by year, respectively.

In addition, the temporal distribution of the selected papers is illustrated in Fig. 1 and Fig. 2. Specifically, Fig. 1 shows the temporal distribution across the three main themes, while Fig. 2 highlights the temporal distribution within the various fields related to trust and reputation systems. Interesting is to observe that Trust and Reputation Systems have encountered significant interest mainly in the last decade, while the remaining topics have encountered a low, but regular, interest over the 25 years of WOA meetings.

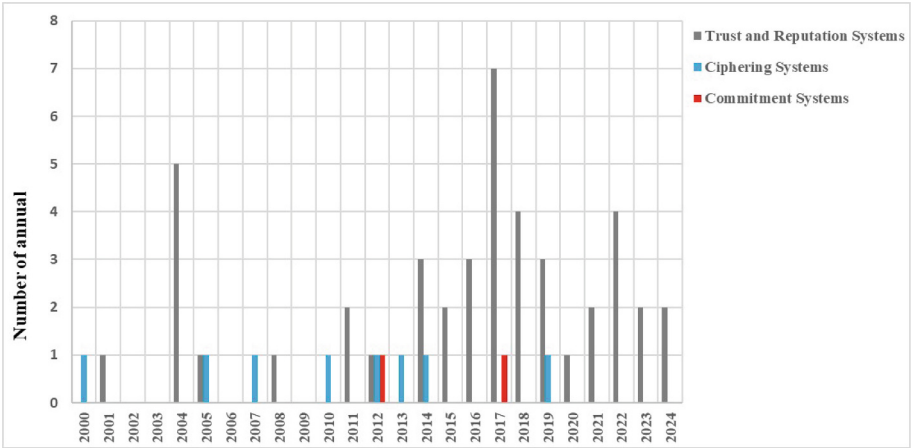


Fig. 1. Contributions in the main themes over the years.

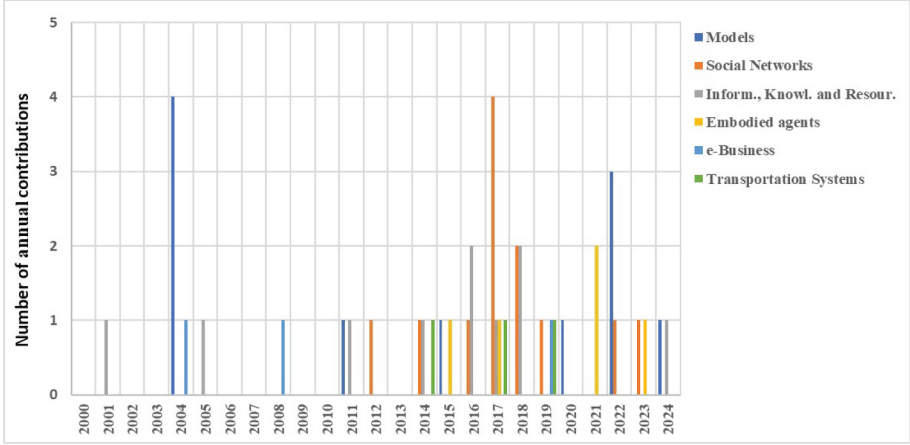


Fig. 2. Trust and Reputation Systems contributions for fields over the years.

3 Trust and Reputation Systems

In choosing an agent as a partner its perceived trustworthiness is essential to make such social interactions as comfortable and fruitful as possible. Consequently, in MAS the concepts of trust, reputation and reliability have long been adopted to optimize the process of partners choice [19]. It is a fact that in the MAS field today, such systems are increasingly used in the context of different application domains in which software agents, also distributed on a large scale, interact and cooperate with other agents.

In particular, to the best of our interest, we can consider trustworthiness as: “the quantified belief by a trustor with respect to the competence, honesty, security and dependability of a trustee within a specified context” [70]. However, it is important to note that relationships founded on trust, reputation, and reliability are typically multidimensional, as they involve varying perspectives that can shift in their practical use depending on the purpose and context of the interaction. For example, in this sense some common dimensions are competence, honesty, safety, trustworthiness, and many others [72].

More specifically, whereas reliability is a subjective measure of trust directly perceived by a trustor with respect to a trustee, reputation represents the measure of trust that an entire community perceives with respect to a specific trustee [74]. As an aggregator of others’ opinions, reputation plays an essential role when a trustor has no knowledge at all, or it is partial and insufficient, with respect to a specific trustee to decide whether to interact with it or not [114]. However, frequently measures of reliability and reputation are combined together into a single measure of trust and a crucial point, in the practical use of these two measures is the choice of the most appropriate way in which to combine them together. In the context of MAS, trust, reliability and reputation (TRR) systems are widely used and there exist a rich literature and surveys on this matter to

which the interested reader may refer to [18, 69, 72, 98, 103, 116]. In this scenario, numerous contributions at WOA emphasized trust and reputation as key criteria for selecting suitable partners.

3.1 Models

Proposals that appear applicable in different application contexts and thus emerge to be of more general use will be presented below.

Starting both from the potential overwhelming number of relationships existing among agents over the Internet and considering that authentication-based approaches may be inadequate (and in any case they do not provide information about agents' skills), Cabri et al. describe in [22] a preliminary approach for evaluating trust in agent interactions, taking into account the specific role played by agents in a generic mobile scenario. Furthermore, since agents do not evaluate trust in the same way as humans, a computational approach based on an analytical formulation that composes several appropriately weighted components was adopted. A set of Java classes was developed to implement the proposed approach.

In several reliability-reputation models, the measures of reliability and reputation are combined together into a synthetic trust measure to support agents' decisions. In [113] the authors present a trust model that considers, from a mathematical point of view, the interdependence between reliability and reputation measures. In other words, a parameter is dynamically computed to determine the relevance of reliability versus reputation based on the agent's direct knowledge about its current counterpart. In addition, the same authors introduce in [113] a new mechanism for calculating reputation, in which the reputation perceived by a trustor agent on a trustee agent is based on the overall trust that every other agent in the MAS has in that trustee agent. This approach requires each agent to solve a linear system. Note that this model is applicable only in MASs where agents are cooperative and share their trust measures with each other. Other researches to assess trust in selecting partners in large agent communities for collaborative processes exploiting different information sources are presented in [86, 124]. Both these studies override the problem of assessing the huge number of possible partners by exploiting agent categories (i.e. reputation classes) by assuming that members of a category that possess similar characteristics then will have similar performances about the same task (with a certain degree of uncertainty). The larger the population and ignorance about the trustworthiness of each individual, the more valuable the role of categories to trust a partner. Both these studies highlight the advantage of assessing trust based on class reputation versus reliability or reputation information to determine the possibility of anticipating a trust score of unknown agents.

The study presented in [108] investigates on group polarization/depolarization, two critical processes underlying relevant behaviors. In other words, polarization processes occur when agents within a community or system interact with each other and gradually move away from their initial views or positions, tending to cluster around opposite poles or, conversely, opposing positions

tend to converge toward a common one (i.e., depolarization). This phenomenon can emerge in a wide range of contexts. In that study, an attempt provides to clarify the role played by similarity and trust, as well as their combination (i.e., the measure of compactness [39]) in various percentages, in different behaviors for modeling the processes of dynamic polarization and proximity depolarization. To this end, a multi-agent community has been simulated, where the software agents replicated salient features of human behaviors and the relationships among the agents were assumed to be qualitatively similar to those occurring in human societies. In addition, the agents also considered the presence of other competing or cooperating agents in planning and implementing possible strategies to achieve their goals.

An operational semantics for communication is described in [130] for the DALI language in which each location is likened to a move in a game, to which other agents can respond with other moves, following a given protocol. Each location returns information by influencing the state of the receiving agent, while other agents must reason about both this expression and their degree of belief, trust, and confidence to assign to the counterpart. In practice, this type of formalization provides a communication architecture that realizes a filter layer in which an agent can describe its mental attitudes and its assumptions about other agents. Note that this paper does not explicitly propose a trust model, although the protocol explicitly considers it.

In an Internet of Things (IoT) context, the paper [51] introduces a theoretical framework for developing and regulating the autonomy of IoT devices with respect to the user they interface with. Indeed, it is critical to understand how the device should behave to stimulate user trust and make the user gradually willing to grant more autonomy to IoT devices to provide better performance for him/her. Specifically, the devices represent the actors of this autonomy, while the delegators are the users. In this scenario, two different forms of autonomy (i.e., execution and decision, the latter modeled on multiple levels) were considered to represent human-IoT devices interactions. On the other hand, the reliability of a device is modeled as a vector of 5 elements, precisely to assess its ability at each level of autonomy. The results of this investigation have clarified the various relationships existing between trust, execution, and decision autonomy by suggesting what approach is suitable to be used to improve the levels of autonomy at which IoT devices can operate.

When the tasks assigned to agents go beyond their original purpose or agent abilities are limited, collaboration between virtual and human agents can be crucial to reach the desired goals. In this context, Falcone and Sapienza in [52] consider cognitive agents able to construct a representation of their own capabilities and those of others (so that they can predict reactions and/or responses), and this plays a key role in social interactions. In addition, identifying possible dependencies to achieve a goal allows agents to discover potential weaknesses or risks. To study the dynamics of dependency networks, a simulation of a block world (in which blocks have different characteristics in terms of shape, color and weight) was conducted. The simulation results confirm that the effect of

dependency is very significant in agent interactions and that, with unreliable partners, better performances are achieved in situations where resources are not available, while in contexts where dependency use is limited, trust assumes a more prominent role.

Maintaining a high level of consistency in a Blockchain Distributed Ledger depends on two key factors: limiting the number of neighbor agents per node and ensuring low delivery times between nodes. Additionally, the absence of unreliable or malicious agents is often crucial, depending on the specific blockchain system. In the past, the authors propose in [61] an algorithm called Trust-based Optimum Neighbor Selection (TONS), designed to: *i*) enable nodes to communicate with a globally optimized selection of reliable and trustworthy neighbors, and, based on trust measures, *ii*) identify misbehaving nodes while optimizing neighbor selection based on delivery times and reputation scores. At the latest WOA workshop, in [62] the same authors present an improved version of TONS, named TONS2. This version introduces a heuristic strategy that reduces the computational overhead by avoiding the need to recalculate a new set of neighbors for each block validation. Simulation results demonstrated that TONS2 is more resource-efficient than its predecessor while maintaining a high quality of neighbor selection. This improvement is particularly beneficial for IoT devices, which often operate with constrained computational and energy resources.

Finally, Ricci and Omicini in [105] discuss on agent coordination artifacts and contexts and how they can play an effective role in shaping trust with regard to social trust, trust in societies, constructive trust, and organizational issues. In more detail, coordination artifacts model the role of abstractions given by the infrastructure with adequate expressiveness and effectiveness to build trust-based strategies, for example, to support the observation and tracking of agents' actions and interactions. In this structured environment, the notion of coordination context can be exploited to model both the interactions and the related events of an agent in its lifetime.

3.2 Social Networks

Online Social Networks (OSNs) enable members to connect and interact with each other over the Internet. These networks play a significant role in modern society, facilitating communication, sharing of information, and the formation of social connections among individuals or groups [67, 115, 126].

In the OSN context, the formation and evolution of groups is a fascinating topic because of its implications at different conceptual levels. Inspired by [39], several works study this topic on the basis of appropriately combined measures of trust and similarity to suggest a social group to an agent and, conversely, to recommend an agent to a group. In this scenario, group homogeneity is critical in determining the satisfaction of group members, and the stability of homogeneity over time is studied in [38] and [87], respectively. Another relevant aspect of OSN is the group cohesion, Fotia et al. investigate in [65] the effects of using local rather than global measures of trust were also considered. The latter topic was also taken up in [80], to measure the overall effectiveness of the groups thus

formed. A practical applications of these studies can be found in [57] and [34]. The former deals with the choice of partners to cooperate in a cloud of things environment. Specifically, groups are formed by using a voting mechanism, in which each vote combines measures of trustworthiness and local reputation, while to form groups with a high level of mutual trust among its members, a distributed algorithm for group formation was designed, optimizing efficiency and effectiveness. In [34] Comi and Rosaci describe SMARTSAN, a recommender system for smart city applications, structured as a multilayer architecture that includes both a P2P platform and a network of social agents, and built on an IoT environment. In SMARTSAN, each social agent is associated with an IoT user or object, and agents can be linked together by trust relationships. Groups of social agents are dynamically formed in SMARTSAN based on measures of both trust and similarity, where trust measures in turn take into account measures of reliability and reputation. Agent groups change over time following changes in users' desires and needs.

Student interactions in OSNs are supported in [40] by means of a MAS exploiting behavioral and trust measures to improve the composition of e-learning student classes. Firstly, behavioral measures refer to a student's proficiency with respect to a set of topics of interest to give educational homogeneity to the class for balancing "offers" and "requests" of support. Trust measures are calculated by combining reliability and reputation measures. Specifically, reputation is based on students' feedback, each of which refers to past interactions, quality of interactions, and historical attitude toward interactions with peers. In turn, these two measures are combined into a single measure called "convenience" and this measure is used to evaluate how convenient it is for a student to join with a class.

In the presence of OSNs organized into thematic groups, trust can play a key role in helping consumers to overcome risk perception. In [63,64] the author assumes that users can be helped by personal software agents in supporting product purchases. Specific trust-based models integrating reliability and reputation have been developed to identify trustworthy recommendations in virtual communities in the presence of groups. The results show that considering only reputation measures underperformed agents who had a direct interaction with another agent (i.e., reliability measures), while the combination of these two measures was found to be the most profitable approach in an online e-Commerce community.

In communities of social agents, nontrivial forms of interaction and cooperation among agents can be realized, and the choice of qualified partners is crucial to avoid being exposed to malicious behaviors. A well known mechanism to identify malicious partners is provided by the EigenTrust algorithm [75], which is a reputation management system designed to compute trust values in peer-to-peer networks. It assigns each peer a global trust score based on the aggregation of local trust values, which are derived from direct interactions. By leveraging transitive trust and iterative calculations, EigenTrust reduces malicious behavior and ensures reliable partner selection in decentralized environments. Unfortunately,

this algorithm has as a side effect that in identifying malicious peers it lowers the reputation of even honest agents. To solve this problem Marcinó et al. propose in [82] a method to preliminary identify the most promising candidates to be considered as malicious and a reputation model that can accurately identify, one at time, malicious agents without introducing the side effects on reputation scores of honest ones typical of the EigenTrust algorithm. This proposal was extended by Marcianò et al. in [83] to identify multiple groups of colluding agents operating independently and simultaneously.

3.3 Information, Knowledge and Resources

In [20], the authors exploit a specific knowledge representation, semantic properties and adaptive algorithms to identify which agents are the most promising candidates to be suggested for a fruitful cooperation. Moreover, each user choice induces changes in the associated knowledge representation, so that the new computed suggestions will be always closer to the user's expectations. Note that the weight associated with each agent can be considered like reputation measures. An extended version of [20] can be found in [21].

ELSA (E-Learning Student Assistant) [112] is an agent-based recommender system assisting a community of students with suggestions about multimedia contents. In this MAS, each student's device hosts a local agent that monitors his/her behavior to build a local profile storing information on the accessed multimedia resources. This profile is then used by an assistant agent to build a student global profile. Based on their global profiles, students are affiliated with one or more partitions where each associated partition agent pre-compiles customized collaborative filtering recommendations taking into account for each student those most similar and those considered more trusted in providing high quality resources. In computing trusted students ELSA, exploits a trust system, derived from that in [114], that dynamically merges reliability and reputation into a global trust measure on the basis of the number of interactions that have occurred in the past.

In [10] Bergenti proposes a MAS where agents' interactions ensure a desired level of security, trust, and the minimum possible number of trusted parties involved. The proposal introduces *Validation-Oriented Ontologies* and *Guarantors* abstractions. To handle the trust aspects, a probabilistic model is adopted, which takes into account a number of parameters, namely: the probability of correctness of information; the probability of the agent fulfilling contractual obligations; the level of confidence placed in the trustee agent; the lower and upper bounds of the probability distribution function of trust. In addition, reward, investment, and penalty are considered in a subjective perspective.

A benefit of using agents is that they can autonomously improve both knowledge and performance through learning. An evolutionary strategy is described in [110], here the concept of cloning, to generate more effective recommendations, is exploited. Specifically, users are served by a set of agents and can replace unsatisfactory ones with others reputed more performing and associated with those users having similar interests. This reputation-driven approach supports

an evolutionary behavior in the agent community, allowing to the better agents to predominate over less productive ones. In this way, individual improvements are reflected in the entire agent community. In addition, to increase the rate of change in the agent population, a proactive mechanism, in which the system independently identifies and suggests to a user, who can accept or reject the suggestion, is proposed. Extensions of [110] can be found in [109, 111].

In the paper [123], the nature of trust in information sources is studied. In particular, the components that influence trust are identified and analyzed, as well as the dependence between the degree of trust in an information and that in the sources that generated it. Based on this analysis, a fuzzy-based computational approach has been designed. The study in [120] describe a tool founded on a theoretical and computational model that analyzes all possible cognitive variables that may influence the level of trust placed on an information source is described. Trust is used to assign different weights to a source in order to assess its credibility, while information is represented by using the Bayesian theory. This tool can be used in different contexts and within different applications. The same authors analyze in [49] the effects of subjective impulsivity by simulating a population of citizens (modeled through cognitive agents) handling weather information. In this situation, agents can act rationally or impulsively (by emulating their neighbors). The effects of impulsivity were shown to be non-homogeneous across the population, i.e., effects at various levels of positivity and negativity. The authors have shown that the social trust tends to have a positive impact on agents with less information and vice versa increases the correctness of the information the agents possess. In that paper, the adopted trust model is an adaptation of a cognitive confidence model [28] and, similarly to the above cited [49], information is represented as a Bayesian probability density function. The same trust model was also applied in [50], to provide a quantitative estimation of the effects of alarmism on the population in the case of hydrogeologic risks. The results show that the overestimation of risks by public authorities decreases trust in them and, as a consequence, citizens tend not to trust a reliable source. Another effect concerns the quantification of damage, so the more citizens are wrong, the more the population will not respond effectively in the case of real events. Recently, in the context of partner selection, in [122] Sapienza and Falcone explore the role of knowledge within dependence networks. Specifically, agents with limited awareness of their environment and potential partners must acquire an adequate level of knowledge about them. This learning process entails cognitive and computational costs due to the continuous updating of agents' beliefs and strategies, which form the basis of their commitments. The experiments conducted highlight the critical role of knowledge, showing that agents with limited knowledge encounter significant challenges, such as reduced collaboration opportunities, inefficient resource utilization, and limited success in achieving their goals. It is important to note, however, that acquiring knowledge comes with costs, particularly when it involves evaluating trustworthiness, which plays a pivotal role in optimizing agent interactions.

A distributed reputation model, operating within an agent-based framework, to ensure the trustworthiness of information sources in vehicle-to-vehicle interactions is presented in [125]. In detail, a reputation model detects malicious and fraudulent sources of information in order to identify the most reliable actors as sources of useful information.

Finally, to maximize the perceived global utility (i.e., QoS) in large-scale federated computing infrastructures (FCI) a trust-based MAS is proposed in [32]. Here, each FCI node (e.g., a grid computing element, a grid site, or a part of a cloud center) is associated with an agent, which manages a set of infrastructural and resource information related to its node. To satisfy complex requirements, the formation of coalitions among FCI, which are chosen on the basis of trust measures combining reliability and reputation measures, is promoted, quantifying the trustworthiness of their peers and used by an algorithm to form groups and friendships among agents (FGF) [33] and able to maximize the overall utility of the entire federation.

3.4 Embodied Agents

Activities including humans and embodied agents, such as robots or physical devices, can benefit from the adoption of TRR systems in partner selection processes in order to optimize the effectiveness and efficiency of their activities.

To this purpose, some aspects related to the trust and delegation tasks in human-robot interactions have been explored in recent editions of WOA in [25–27]. In trust theory, the principle of delegation [29] is considered one of its founding principles. In particular, cooperation and the mental attitude of delegation has been explored by Cantucci et al. in their research [25], where a declarative, knowledge-oriented, plan-based computational model, implemented on a JaCaMo Framework [17] is designed. This computational model integrates the concept of adjustable social autonomy as a basis for an effective human-robot interaction. However, each request of collaboration also poses an inherent risk of collaborative conflict between the human (the trustor) and the robot (the trustee). To investigate this problem, in [27] a cognitive architecture for profitable human-robot collaboration based on some theoretical principles (i.e., adoption and delegation, mind, adjustable social autonomy, and trust theories) is presented. Based on a Belief-Desire-Intention (BDI) model [104], this architecture enables robots with their own mental states to exhibit a wide range of cognitive abilities able to maximize user performance evaluation. Furthermore, in the proposal presented in [26] a cognitive approach to model human-robot interactions pays attention to four attributes, namely: usefulness, effectiveness, acceptability and trustworthiness. Specifically, the mental attitude of trust is related to the type of cooperation provided by the designed robot and is the mental counterpart of delegation. This means that the robot must be able to achieve the user's goals or needs, but it must also be able to delegate tasks, based on an evaluation of other agents (artificial or human) potentially involved in the interaction.

A proposal about users' acceptance of IoT systems is discussed in [121] adopting a model, also considering security and privacy issues on the user side, by

adopting both the concepts of trust and control, suitably ruled by feedback about behaviors and performed actions. Fortino et al. present in [58], a trust-based model simulating an industrial scenario which optimizing the efficiency and effectiveness of groups of workers and autonomous guided vehicles (AGVs). The simulation shows how the adoption of trust in group formation processes can optimize processing time and production quality.

Robots are embodied agents and, in order to have fruitful interactions with humans, it is essential that they be perceived as potential partners with an appropriate level of trustworthiness. The study presented in [30], a study investigates this in relation to a virtual agent using three different interaction conditions (e.g., a mobile application, a humanoid robot using speech only, and a humanoid robot using both speech and gestures) to provide movie recommendations.

3.5 e-Business

The term e-Business refers to all those economic activities carried out by exploiting ITC tools. Just like traditional business activities, which require reliable counterparts to perform economically valuable tasks, it is even more intuitive that this holds true when such activities are conducted over the Internet. In this case, TRR systems can help to create a positive atmosphere in the execution of business activities by providing information about agents based on their past behaviors.

Applications of TRR systems to e-business were explored at WOA in [14, 90]. More in detail, in [14] Bergenti et al. describe a framework for exchanging customer information among e-commerce applications. A Trust Management System, leveraging stated trust relationships, supports personalized information sharing in order to select the best service providers based on their characteristics. More generally, service providers are supported in the sharing of information from a trusted third party (i.e., the User Modeling Agent) without the need to modify their individual representation models or, alternatively if they do not have their own representation model, requesting the support of the User Modeling Agent. On the other hand, an external reputation system is considered in [90] as a key element for an auction bidding process to enable an agent as a bidder, but without deepening this issue.

3.6 Transportation Systems

In the field of transportation systems, few applications of trust and reputation to qualify a counterpart were presented at the WOA events. For example, reputation is exploited in [100] to estimate the accuracy of measurements made by nodes of a sensor grid that monitor traffic parameters, and in this way, errors made by the sensors can be mitigated.

Car-sharing and car-pooling mobility, which can complement transit systems, can also benefit from the adoption of TRR systems. For instance, a trust-based MAS is adopted in a peer-to-peer car-sharing scenario [97] to encourage and support private owners to share their cars. In this study, the agents monitor

the driving habits of car-sharing clients and assist them in improving their driving. Over time, the agents also build driver's reputation measures that will be used by car-owners for avoiding to rent their car to unqualified drivers and to customized rent rates awarding the best drivers. Car-pooling is another form of shared mobility that can optimize the load factor of the car, although the origin and destination of the trip must be the same for all the passengers. In this respect, in [101] the authors design an on-demand service that adopts variable rates based on the length of the trip and the number of participants. Multi-agent, reputation (to select good travel mates) and blockchain technologies are used together, and a suitable dynamic routing algorithm has been developed.

4 Ciphering Systems

In choosing the agent to interact with, the use of encryption offers a number of advantages and limitations that affect its usefulness and applicability in different contexts [54, 68]. Advantages include: privacy, confidentiality and data integrity; authentication of the parties involved; protection from security breaches; and regulatory compliance in the presence of regulations requiring protection of sensitive data. In contrast, the main limitations in adopting cryptography lie in: increased implementation complexity (with likely costs in terms of hardware, software and personnel training); possible vulnerabilities present in the cryptographic protocols themselves or in the systems that implement them; complexity in key management; and risks of security compromise in the case of key loss.

By comparing trust and reputation systems with cryptographic systems, both tools are used to assess the reliability and security of an environment, but they are based on different concepts and approaches [137]. Trust and Reputation systems are characterized by assessing trustworthiness by leveraging on users' past experience. This realizes a social evaluation mechanism based on users' shared experiences and the reputation is accumulated by an individual or entity over time [129]. Therefore, the trustworthiness of potential partners is determined by the aggregate evaluation of many users, also in order to mitigate the effects of subjectivity, that is the influence that users' subjective opinions give that may vary based on their personal experiences [15]. In addition, trust and reputation systems are adaptable over time based on changing circumstances, user interactions, and manipulation attacks such as false feedback or fraudulent ratings [89].

Encryption systems are completely different because they are based on mathematical algorithms and protocols whose robustness and adequacy realize the effect "trust". Therefore, evaluations do not involve any social aspects, but only require the proper management of every cryptographic component. However, the "aseptic objectivity" provided by cryptographic systems is not synonymous of a superiority over TRR systems.

In summary, while trust and reputation systems are based on past users' experience and social evaluation to foresee expected future behaviors, cryptographic systems offer reliability, confidentiality, resilience, and security based on mathematical and management principles. Each of the two approaches has

advantages and limitations, and both can be used to improve security and trust in an online environment. However, the combination of these two technologies is able to provide a greater effectiveness with respect to the single approaches, although at the cost of significantly increasing the complexity of the system.

Among the paper presented at the WOA that in [107], presented by Roggero et al., propose an automated trust negotiation mechanism for MASs, where the term trusted refers only to a certified and verified identity of the agent acting in the Internet. Specifically, this paper describes a protocol for the creation of trusted and untamperable credentials, issued by a third-party entity, in order to evaluate an access request, verifying both the possession of attributes and prerequisites. Conceptually similar to that in [107], Tomaiolo and Agazzi describe in [1] a trust negotiation framework for Web services based on peer-to-peer trust relationships. This framework allows users to build trust automatically by incrementally revealing credentials. It includes several components to handle a range of format conversions for messages, policies, and credentials, ensuring interoperability among services. Always, in a Web Service context, the authors of [131] discuss a cross-domain security approach, using cryptography but without leveraging centralized or hierarchical certificate authorities or public lists of names. This work adopts a peer-to-peer delegation mechanism. The results are a library for issuing and verifying delegation chains, a security service with a SOAP [71] interface, a security service with a RESTFUL [106] interface, and other components suitable for an open environment.

On the same mechanisms [13, 66] present decentralized MASs for OSN, called Blogracy, deals with social information retrieval issues. It introduces a key-based identity system and a model of social relations for distributing resources. At a lower level, Blogracy uses widespread and stable peer-to-peer technologies, such as distributed hash tables [85] and BitTorrent [102]. The most salient features of Blogracy are: *i*) anonymity and resilience to censorship; *ii*) authenticated content; *iii*) semantic interoperability; and *iv*) data availability. The same Blogracy platform is also used in [3], combined with a spanning tree, to spread online notifications on a distributed OSN to solve privacy, censorship and personal data control issues. The spanning tree robustness is ensured by connection randomness to join with a group, while in the case of node failures a recursive election procedure is implemented for the tree reconstruction.

The adoption of security assertions to improve existing security systems in federated contexts poses new problems that are addressed in [99]. This paper leverages conventional Public-Key Infrastructure (PKI) [95] to witness a web of trusted relationships in open, decentralized agent-based environments. It enables direct interoperability between parties by controlling the flow of authorizations in a XML-based chain of multiple certificates within a trusted delegation chain.

That in [8] describes a mobile agent framework called Secure and Open Mobile Agent (SOMA). SOMA provides several solutions for security and interoperability, two key aspects in mobile environments to avoid of interacting with malicious agents. In this context, authentication and authorization processes (based on credentials) are essential. In particular, SOMA complies with the

OMG/CORBA [47] and MASIF [88] standards to which the management of privacy and security instances using cryptographic techniques is delegated.

5 Commitment Systems

In MASs, commitment-based partner selection is an area of research that focuses on the selection of partners or allies by agents based on the principle that potential agent partners will keep its commitments [31]. This is a key consideration in many situations where agents must collaborate or interact with each other in a distributed context.

Commitment approaches that can be implemented in partners selection include the history of interactions and reputation of partners (peculiar elements also shared with trust and reputation systems), the risks associated within a possible collaboration with a given partner (considering the probability that they will not keep their commitments and the negative consequences of this), and their stated commitment, provide a basis for assessing their trustworthiness.

However, even in systems based on commitment, it must be taken into account that the relationships between agents and their trustworthiness may vary over time [127]. Agents must balance the risk of working with untrustworthy partners against the potential benefits of the collaboration itself, and it is important that information on reputation and stated commitments be transparent and easily accessible to the agents to enable them to make informed decisions on partners selection.

The main feature of commitment protocols is their declarative nature, which allows one to specify a “commitment to do something” and whether what is promised is accomplished. In the position paper [4], agent coordination, infrastructure, event observability, and composition of coordination patterns are analyzed in light of a commitment process. Thus, commitment protocols enable each agent to know its potential counterparts and make informed choices in terms of making and fulfilling commitments.

In terms of commitment, complex activities may involve multiple parties that require each actor to fulfill commitments from a coordination and interaction perspective. To address the lack of business artifacts, a MAS approach is proposed in [6] to support the ideas both that the services for which business artifacts can be operated should be encapsulated and organized into goal-oriented containers, and that a normative layer should be introduced to capture the expected behaviors of the parties. This work could involve three main aspects, namely: an explicit normative layer, understanding how agents could plan the use of business artifacts to achieve their goals, and finally, standardizing the commitment life-cycle.

6 Conclusion

Choosing the best partners is a major strategic issue in a MAS scenario (where each agent has its own set of skills, experience and behaviors), and collaboration

is an essential element in ensuring that each agent achieves its own objectives by solving from the simplest to the most complex tasks. Carrying out correct partners selection processes has a deep impact on the entire agent community, since suitable partners improve confidence between parties and lead to increasing the social capital of MAS. A number of challenges need to be addressed to solve this problem, and several techniques have been proposed in the literature for this purpose. Each of them aims to build fruitful relationships among agents and to create an atmosphere of confidence that can reduce information asymmetry among agents and marginalize malicious agents.

In the above context, this chapter provided a comprehensive overview about the scientific contributions presented at the annual WOA workshops from 2000 to 2024, focusing on the problem of selecting one or more counterparts for interaction and cooperation. This selection process is based on key criteria such as trust, reliability, reputation, encryption, and commitment across multiple domains. Over the span of 25 years, this critical issue has been consistently addressed by WOA authors in 53 papers included in this analysis, out of a total of 515 regular papers presented at WOA. This represents a significant part of the workshop's scientific production, underscoring the centrality of this theme in the WOA community. The prominence of this topic within WOA is further evidenced by the variety of strategies proposed in the papers, reflecting the diverse and innovative approaches taken to tackle the challenges of partner selection. These contributions highlight the dynamic interplay between theoretical advancements and practical applications, spanning fields such as, for instance, agent-based systems, distributed computing, and autonomous decision-making. Moreover, the relevance of these researches extend beyond WOA, as demonstrated by the broader scientific community's recognition of the value of these contributions. In many cases, these papers have been further refined and expanded, resulting in extended versions published in leading journals in the field. Such outcomes highlight the quality and originality of the work presented at WOA and its ability to anticipate and address emerging trends in agent-based research impacting on both wider academic and industrial landscapes. In conclusion, the focus on trust, reliability, reputation, encryption, and commitment within the WOA framework has not only enriched the workshop's bibliography but also contributed meaningfully to the advancement of knowledge in the field of agents. The enduring interest in this topic, as reflected in both the volume of research and its influence on the broader scientific literature, attests to its significance and the pivotal role played by WOA in fostering innovation and collaboration within this domain.

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